



microring modulators.



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- different PN junction strategies in literature.
- Example Simulation.



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- different PN junction strategies in literature.



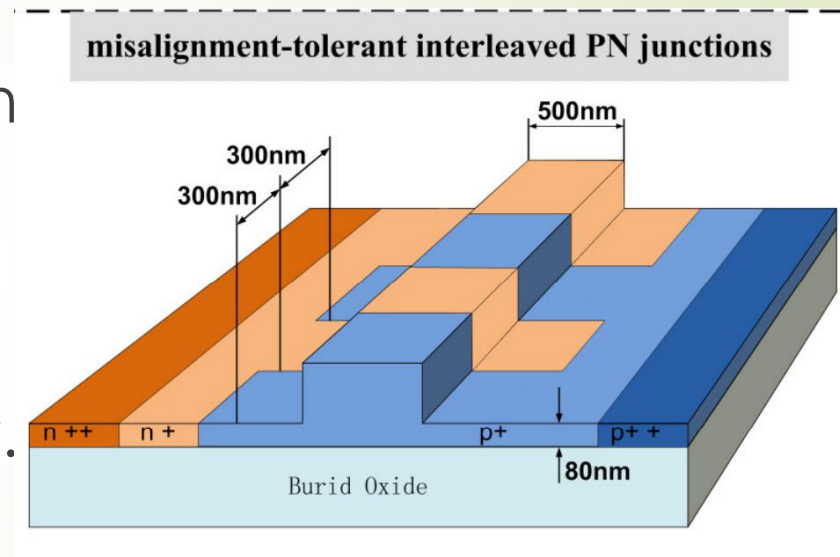
different PN junction strategies in literature.

- **what a PN geometry is trying to optimize?**
- maximize optical overlap → high modulation efficiency.
- minimize capacitance → high RC bandwidth.
- minimize light absorption → low loss.
- maintain fabrication margin → yield & reliability.

➤ different PN junction strategies in literature.

Cascaded ring modulator (Hu et al.)

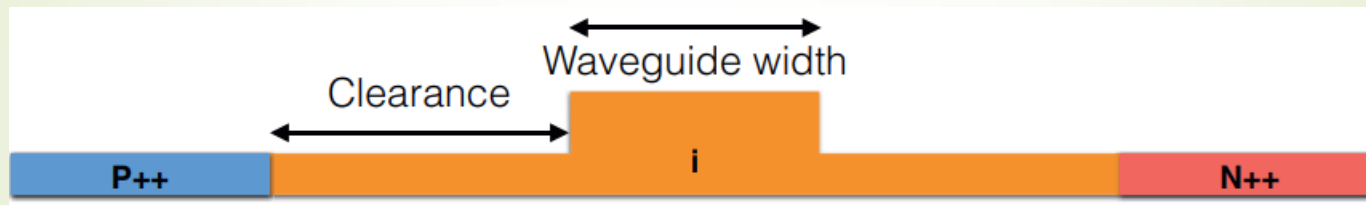
- Interleaved PN 600 nm length (300 nm for both P & N), 700 nm width, @ $2 \times 10^{17} \text{ cm}^{-3}$ doping.
- Higher modulation efficiency.



➡ different PN junction strategies in literature.

Optical peaking design (Müller et al.)

- A reversed biased PIN junction with Nominal concentrations $p^+ = 3 \times 10^{17} \text{ cm}^{-3}$, $N^- = 1 \times 10^{17} \text{ cm}^{-3}$.

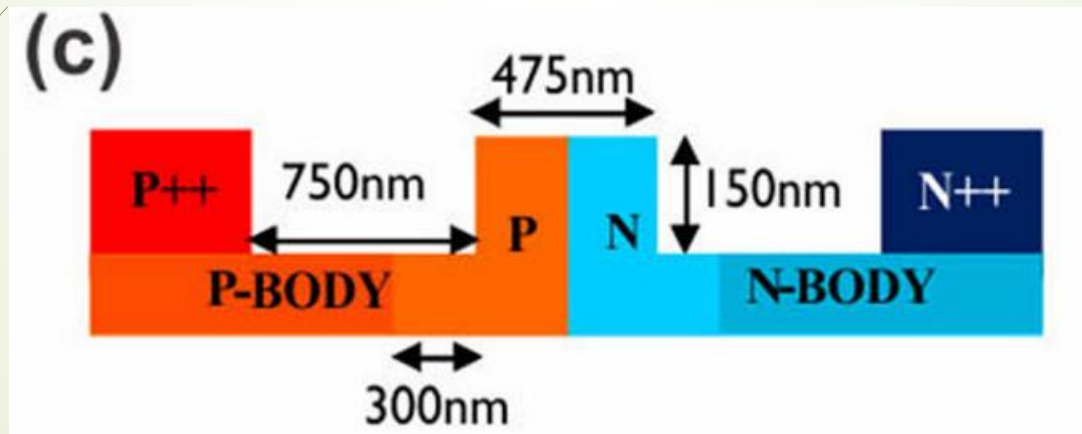


- increased size of the junction by introducing an undoped (intrinsic) region between the P and N regions.
- Lower junction capacitance (C_j).
- Reduced loss.

- different PN junction strategies in literature.

CMOS co-packaged ring (Moralis-Pegios et al.)

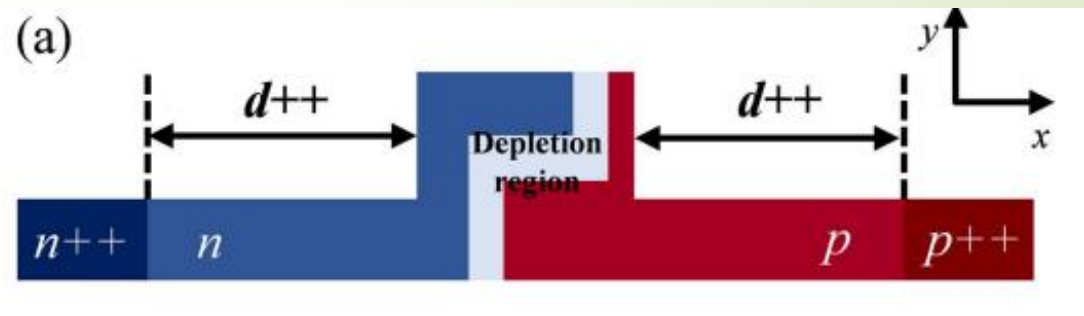
- Lateral PN, highly doped; with a total length of $47.1\mu\text{m}$.



➡ different PN junction strategies in literature.

Lateral zigzag PN (Cai et al.)

- lateral-zigzag PN junction for silicon modulator working in C and O bands.



- significantly increased modulation efficiency while keeping a low C_j , high RC bandwidth.
- The depletion region is composed of a vertical PN junction in the middle and two horizontal PN junctions on both sides.



contents



Example Simulation.

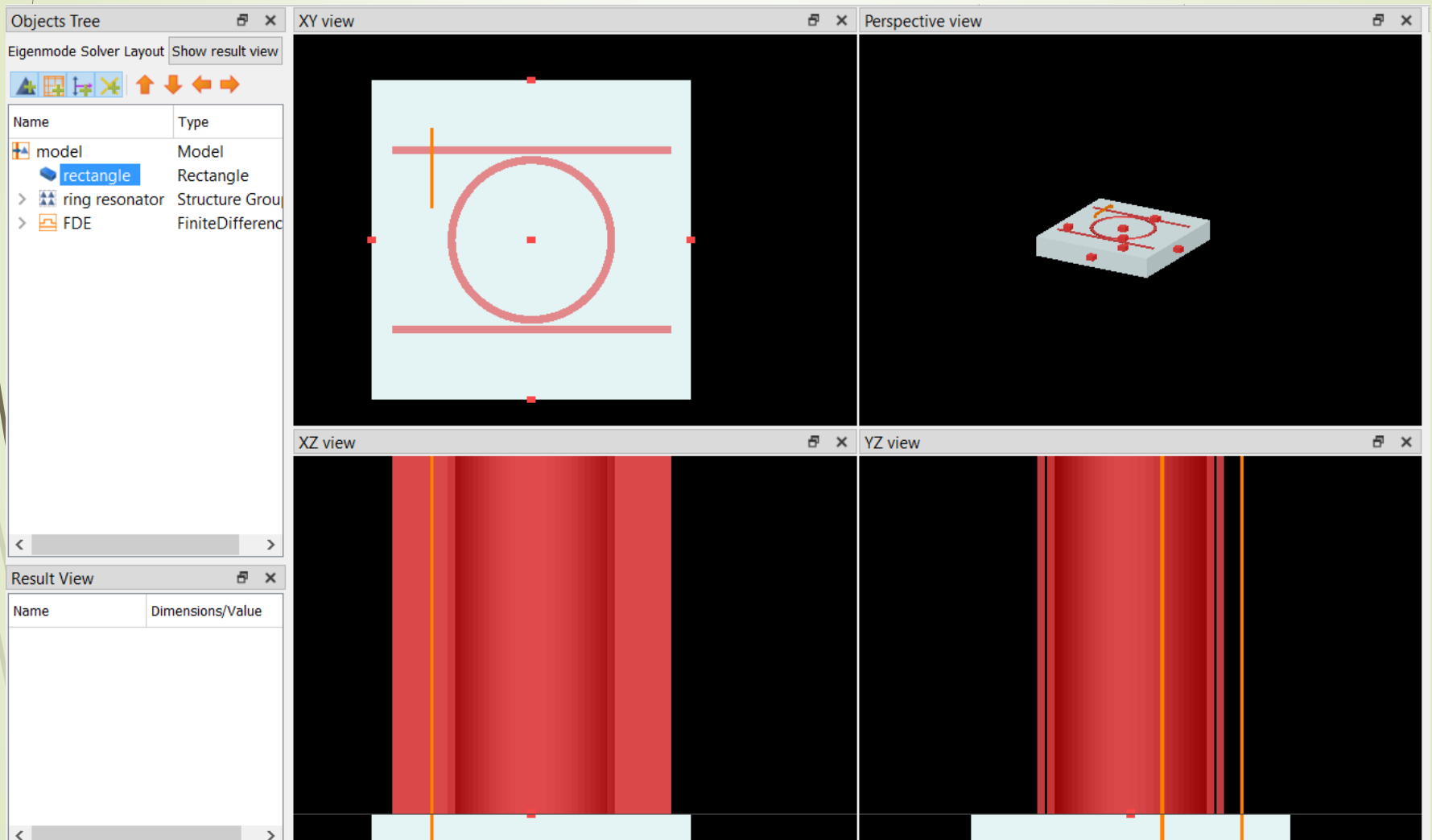




Example Simulation

- used MODE to design and simulate a ring resonator. Free spectral range (FSR) and quality factor (Q factor) are key performance metrics.
- setup the model.
- use the eigenmode solver and the analytical to design the ring resonator.

Example Simulation



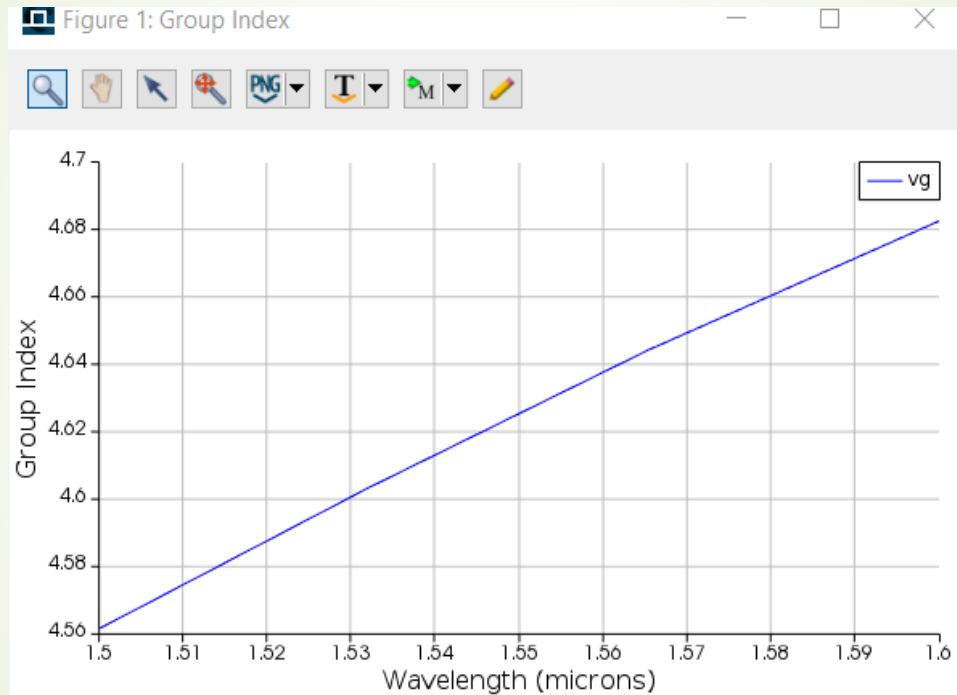
Example Simulation

Eigensolver Analysis

Mode list

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	TE polarization fraction (Ey)	waveguide TE/TM fraction (%)	effective area (μm^2)
1	1.998598	1.5	0.0000	4.561554	95	64.26 / 80.3	0.198428
2	1.436467	1.5	0.0000	1.963465	6	91.5 / 91.29	0.459713
3	1.253154	1.5	0.0000	1.544504	1	85.2 / 95.02	1.27576
4	1.171002	1.5	0.0000	1.667795	3	87.52 / 82.35	1.62388
5	1.072148	1.5	0.0000	1.810784	9	89.84 / 71.8	1.66697
6	1.050281	1.5	0.0000	1.616663	36	92.76 / 87.53	0.876948
7	0.9619466	1.5	0.0000	1.702389	98	98.5 / 58.8	1.25768
8	0.9417443	1.5	0.0000	2.238008	13	85.65 / 59.03	1.93068

Example Simulation



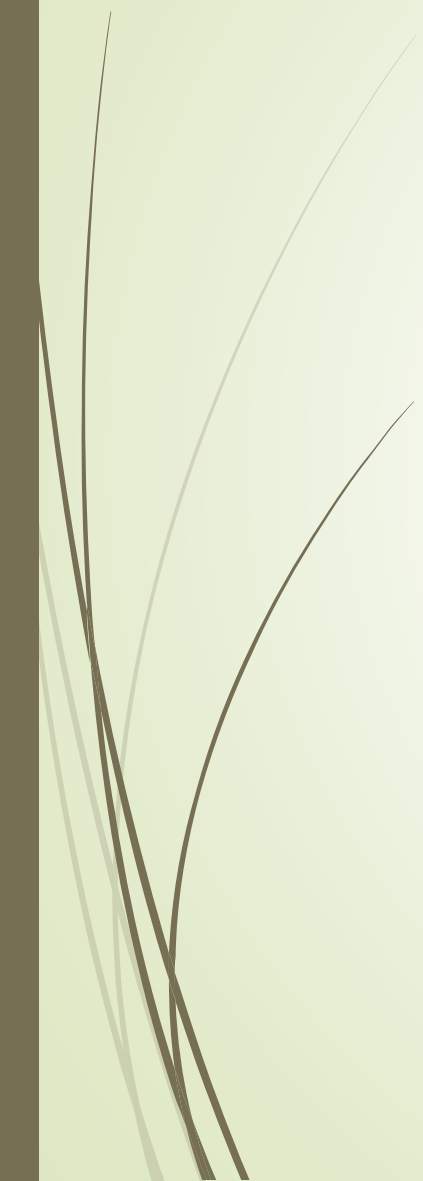
Example Simulation

$$FSR = \frac{\lambda^2}{n_g L}$$

to design a ring resonator for a WDM system with a channel spacing of 200GHz (1.6nm at 1550nm). We want to design the system to drop every 16th channel.

We can see that the group index is approximately 4.63 at 1550nm.

$L = 20.2\mu\text{m}$.



Thanks 😊